

## DYNAMICAL SYSTEMS - MA3081

### Solutions to Exercise Sheet 12

**Exercise 12.1** (Invariant measure of the logistic map).

Show that the measure  $\mu$  given by the density

$$h(x) = \frac{1}{\pi\sqrt{x(1-x)}}$$

with respect to the Lebesgue measure on  $[0,1]$  is an invariant measure of the logistic map

$$g: [0,1] \rightarrow [0,1], \quad x \mapsto 4x(1-x).$$

*Solution.* Let  $B \subset [0,1]$  be a Borel set. We have to show that  $\mu(B) = \mu(g^{-1}(B))$ . We have

$$\begin{aligned} \mu(g^{-1}(B)) &= \int_{g^{-1}(B)} h(x) dx \\ &= \int_{g^{-1}(B) \cap [0, 1/2]} h(x) dx + \int_{g^{-1}(B) \cap [1/2, 1]} h(x) dx. \end{aligned}$$

Note that  $g$  is bijective on both of the intervals  $[0, \frac{1}{2}]$  and  $[\frac{1}{2}, 1]$  and the inverse is given by  $\lambda_1: [0,1] \rightarrow [0, \frac{1}{2}]$ ,  $\lambda_1(x) = \frac{1}{2}(1 - \sqrt{1-x})$  and  $\lambda_2: [0,1] \rightarrow [\frac{1}{2}, 1]$ ,  $\lambda_2(x) = \frac{1}{2}(1 + \sqrt{1-x})$  respectively.

This allows to write

$$\mu(g^{-1}(B)) = \int_{\lambda_1(B)} h(x) dx + \int_{\lambda_2(B)} h(x) dx.$$

Since  $g$  and  $h$  are symmetric around  $\frac{1}{2}$ , the two integrals are the same and thus

$$\mu(g^{-1}(B)) = 2 \int_{\lambda_1(B)} h(x) dx, .$$

As  $\lambda_1$  is a diffeomorphism, we can transform this integral to

$$\mu(g^{-1}(B)) = 2 \int_{\lambda_1(B)} h(x) dx = 2 \int_B h(\lambda_1(y)) \cdot |\lambda_1'(y)| dy.$$

It thus suffices to verify that  $2h(\lambda_1(y))|\lambda_1'(y)| = h(y)$  in order to show equality of  $\mu$  and  $g_*\mu$ .

With  $\lambda_1'(y) = \frac{1}{4\sqrt{1-y}}$  we compute

$$\begin{aligned} 2h(\lambda_1(y))|\lambda_1'(y)| &= \frac{2}{\pi\sqrt{\frac{1}{2}(1-\sqrt{1-y})(1-\frac{1}{2}(1-\sqrt{1-y}))}} \frac{1}{4\sqrt{1-y}} \\ &= \frac{2}{\pi\sqrt{\frac{1}{2}(1-\sqrt{1-y})(\frac{1}{2}+\frac{1}{2}\sqrt{1-y})}} \frac{1}{4\sqrt{1-y}} \\ &= \frac{2}{\pi\frac{1}{2}\sqrt{(1-(1-y))}} \frac{1}{4\sqrt{1-y}} = \frac{1}{\pi\sqrt{y}\sqrt{1-y}} = h(y) \end{aligned}$$

Remark: The same arguments also work with intervals (instead of general Borel sets) and the classical substitution formula for integration.  $\square$

**Exercise 12.2** (Invariant measures of the doubling map).

Consider the doubling map

$$g: [0, 1] \rightarrow [0, 1], \quad g(x) = 2x \mod 1.$$

Find three invariant measures of  $g$ .

*Hint:* Dirac measures.

*Solution.*

- The Dirac measure  $\mu = \delta_0$  at the equilibrium point 0, i.e.

$$\delta_0(A) = \begin{cases} 1 & \text{if } 0 \in A \\ 0 & \text{if } 0 \notin A \end{cases}$$

is an invariant measure since  $0 \in A \Leftrightarrow 0 \in g^{-1}(A)$  and thus  $\mu(A) = 1 \Leftrightarrow \mu(g^{-1}(A)) = 1$ .

- The averaged Dirac measure at a periodic orbit is also an invariant measure. Take for example the 2-periodic orbit  $x_1 = \frac{1}{3}, x_2 = \frac{2}{3}$  and set

$$\mu = \frac{1}{2}\delta_{\frac{1}{3}} + \frac{1}{2}\delta_{\frac{2}{3}}, \text{ i.e. } \mu(A) = \begin{cases} 1 & \text{if } \frac{1}{3} \in A \text{ and } \frac{2}{3} \in A, \\ \frac{1}{2} & \text{if either } \frac{1}{3} \in A \text{ or } \frac{2}{3} \in A, \\ 0 & \text{else,} \end{cases}$$

This measure is invariant since  $\frac{1}{3} \in A \Leftrightarrow \frac{2}{3} \in g^{-1}(A)$  and  $\frac{2}{3} \in A \Leftrightarrow \frac{1}{3} \in g^{-1}(A)$ .

- Also the Lebesgue measure  $\lambda$  is invariant. For any interval  $[a, b] \subset [0, 1]$  we have

$$g^{-1}([a, b]) = [\frac{a}{2}, \frac{b}{2}] \cup [\frac{a}{2} + \frac{1}{2}, \frac{b}{2} + \frac{1}{2}].$$

Hence

$$\lambda(g^{-1}([a, b])) = \frac{b}{2} - \frac{a}{2} + \frac{b}{2} - \frac{a}{2} = b - a = \lambda([a, b]).$$

□

**Exercise 12.3** (More invariant measures).

Consider the map

$$f: [0, 1] \rightarrow [0, 1], \quad f(x) = \begin{cases} 2x & \text{if } x \in [0, \frac{1}{2}] \\ x - \frac{1}{2} & \text{if } x \in (\frac{1}{2}, 1] \end{cases}.$$

Find an invariant measure of  $f$  that is given by a density with respect to the Lebesgue measure.

*Hint:* Look for a piecewise constant density.

*Solution.* We show that the measure  $\mu$  given by a piecewise constant density

$$h(x) = \begin{cases} h_1 & \text{if } x \in [0, \frac{1}{2}] \\ h_2 & \text{if } x \in (\frac{1}{2}, 1] \end{cases}$$

for a suitable choice of  $h_1, h_2 \in \mathbb{R}$  is invariant.

Note that

$$\mathcal{E} = \left\{ [a, b], 0 \leq a \leq b \leq \frac{1}{2} \right\} \cup \left\{ (c, d], \frac{1}{2} \leq c \leq d \leq 1 \right\} \cup \{\emptyset\}$$

is an intersection-closed generator of the Borel- $\sigma$ -algebra on  $[0, 1]$ , so it suffices to check equality of the measures  $\mu$  and  $\mu(f^{-1}(\cdot))$  only for sets  $E \in \mathcal{E}$ .

Take an interval  $[a, b] \subset [0, \frac{1}{2}]$ . Its preimage is given by  $f^{-1}([a, b]) = [\frac{a}{2}, \frac{b}{2}] \cup [a + \frac{1}{2}, b + \frac{1}{2}]$ . This yields

$$\mu(f^{-1}([a, b])) = h_1 \cdot \frac{1}{2}(b - a) + h_2 \cdot (b - a) = \frac{h_1 + 2h_2}{2}(b - a).$$

For an interval  $(c, d] \subset (\frac{1}{2}, 1]$  the preimage is  $f^{-1}((c, d]) = (\frac{c}{2}, \frac{d}{2}] \subset [0, \frac{1}{2}]$ , so that

$$\mu(f^{-1}((c, d])) = \frac{h_1}{2}(d - c)$$

For the above intervals we have  $\mu([a, b]) = h_1(b - a)$  and  $\mu((c, d]) = h_2(d - c)$  so for an invariant measure we have to solve the equations

$$\begin{aligned} \frac{h_1 + 2h_2}{2}(b - a) &= h_1(b - a) \\ \frac{h_1}{2}(d - c) &= h_2(d - c). \end{aligned}$$

We find that these equations do not have a unique solution, but any choice of  $h_1, h_2 \in \mathbb{R}$  with  $h_1 = 2h_2$  solves them. However, we further have the requirement that  $\mu$  is a probability measure, so  $\frac{1}{2}h_1 + \frac{1}{2}h_2 = 1$  needs to be satisfied as well. This yields  $h_1 = \frac{4}{3}$  and  $h_2 = \frac{2}{3}$ .  $\square$